School of Computational Sciences and Artificial Intelligence
DSAI 305-Interpretability & Explainability in AI - Spring 2026
Phase #0-1
Project Proposal

Explainable AI for Early Risk Prediction of Endometriosis, and Systemic Lupus Erythematosus

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1. What is the problem you are solving and its importance? (0.5 points)

1- Problem Statement:

Problem: We are developing an experimental Explainable AI (XAI) framework for the early risk prediction and classification of two clinically significant, female-predominant systemic conditions: Endometriosis and Systemic Lupus Erythematosus (SLE). Both conditions share overlapping and vague clinical presentation, including chronic inflammation, fatigue, joint pain, immune dysregulation, and hormonal irregularities, which notoriously lead to diagnostic delays of 7–10 years globally. While machine learning models can process complex clinical records and biomarker data to identify early risk patterns, these models traditionally operate as black boxes, preventing clinician trust and real world adoption.

By placing Explainability at the core of our framework, our system will transparently reveal the specific physiological markers, antibody profiles, and clinical features that drive risk predictions for each individual patient, transforming AI from a purely predictive algorithm into a clinically trustworthy decision support tool.

Importance: SLE affects mostly women of reproductive age and is frequently misdiagnosed as other inflammatory or rheumatological conditions due to its heterogeneous multisystem presentation. Endometriosis affects approximately 10% of women globally and suffers from an average diagnostic delay of 7–10 years due to symptom overlap with other pelvic and inflammatory conditions. Both diseases share immunological markers, elevated inflammatory proteins, autoantibodies, and complement abnormalities, making early automated differentiation both clinically necessary and technically challenging.

A transparent, interpretable AI framework that can flag high-risk patients from routine blood test panels and antibody profiles and explain exactly which features drive each prediction has the potential to dramatically reduce diagnostic delays, improve early intervention rates, and build genuine clinician trust in AI assisted diagnostics. This project directly addresses the critical gap between high performing but blackbox ML models and the transparency requirements of real world clinical practice.

2. Which algorithms/models do you plan to use/develop? (0.5 points)

2. Algorithms / Models :

We will implement a multi-staged supervised learning pipeline that transitions from high-interpretability baselines to high-performance ensemble architectures, integrated with an XAI (Explainable AI) diagnostic layer.

A. Model Development Pipeline

- Baseline Benchmarking: We will use Logistic Regression and Decision Trees to establish a performance floor. These models offer inherent transparency but may struggle with the non-linear complexities of multi-antibody profiles.
- Tree-Based Boosting (XGBoost & LightGBM): Following the methodology in [1] and [4], we will utilize Gradient Boosting machines. These are particularly effective at handling the sparse, binary nature of our antibody data (52 features) and capturing the non-linear interactions between blood markers like ESR and CRP.
- Stacked Ensemble Architecture: Inspired by [12], we will develop a Stacking Classifier. This will use a diverse set of "base learners" (e.g., Random Forest, SVM, and XGBoost) whose predictions are fed into a meta-learner (Logistic Regression). This approach is proven to improve sensitivity critical for reducing the 7–10 year diagnostic delay in Endometriosis.

B. The Explainability (XAI) Framework

- Global Interpretability (SHAP): We will use SHAP Summary Plots to rank the clinical features that drive risk across the entire population. This

allows us to verify if the model correctly prioritizes "gold standard" markers like Anti-dsDNA for SLE.

- Local Interpretability (SHAP Force Plots & LIME): For individual patient risk assessments, we will generate Force Plots. These visualize how specific biomarkers (e.g., a high "Sickness Duration" or "Low C3") pushed the model toward a positive diagnosis, providing a "reasoning" trace for clinicians.
- Visual Evidence (Grad-CAM): If MRI data is incorporated, we will apply Grad-CAM (Gradient-weighted Class Activation Mapping) to CNN layers. This will highlight the specific anatomical regions in pelvic scans that the model identifies as characteristic of Endometriosis, following the approach in [11].

C. Handling Class Imbalance & Fidelity

Given the extreme imbalance in our source data, the pipeline includes a Synthetic Data Validation step. We will evaluate the models using Precision-Recall (PR) Curves rather than accuracy, ensuring the CTGAN-generated Endometriosis samples do not introduce artifacts that lead to over-optimistic performance.

3. What data will you use (how will you get it)? Give data specifics (size, format, source, etc.) (1 point)

Source: All data was obtained from a single publicly accessible Kaggle repository:

https://www.kaggle.com/datasets/abdullahragheb/all-autoimmune-disorder-10k

Five CSV files were downloaded from this source:

File	Rows	Columns
Autoimmune_Disorder_10k_with_All_Disorders.csv	12,500	79
Comprehensive_Autoimmune_Disorder_Dataset.csv	10,088	79

Complete_Updated_Autoimmune_Disorder_Dataset.csv	13,812	80
Complete_Updated_Autoimmune_Disorder_Dataset2.csv	13,812	79
Final_Balanced_Autoimmune_Disorder_Dataset.csv	11,712	79
Total raw	**61,924**	—

Data Construction Pipeline:

The five files were merged by keeping only the 78 common columns across all files. Patient_ID was dropped as IDs differed across files. All five files were concatenated into 61,924 rows, then exact duplicate rows were removed, leaving 40,389 unique rows saved as TEAM22_Final_Dataset.csv. Upon inspecting class distribution, a critical imbalance was discovered: Endometriosis had only 184 samples (0.46% of the dataset), which would cause any ML model to completely ignore this class. To fix this, CTGAN (Conditional Tabular GAN) was trained for 300 epochs on the 184 real Endometriosis rows, learning their blood count patterns, antibody profiles, and symptom distributions. CTGAN generated 4,816 synthetic Endometriosis rows, which were added to the 40,389 merged rows to produce the final master dataset.

Final Dataset: TEAM22_Master_Dataset.csv

Property	Value
Format	Tabular CSV
Total rows	45,205 patients
Total columns	79 features

Unique diagnoses	118
SLE samples	5,363
Endometriosis samples	5,000 (184 real + 4,816 CTGAN synthetic)
Normal/Healthy controls	6,960
Missing values	0 (median imputation applied)
Augmentation method	CTGAN (300 epochs)

Feature Breakdown (79 columns):

- Demographics (2): Age, Gender
- Continuous lab results (24): RBC Count, Hemoglobin, WBC Count, ESR, CRP, C3, C4, Platelets, Lymphocytes, Neutrophils, Monocytes, Eosinophils, Hematocrit, MCV, MCH, MCHC, RDW, MPV, Basophils, Reticulocyte Count, Esbach, MBL Level, Sickness Duration Months
- Binary antibody tests (52, 0=absent / 1=present): ANA, Anti-dsDNA, Anti-Sm, ACPA, Anti-TPO, Rheumatoid Factor, ANCA, Anti_Ro_SSA, Anti_La_SSB, Anti_RNP, Anti_Jo1, anti_Scl_70 and 40 additional disease-specific antibodies
- Binary symptoms (10): Fatigue, Joint pain, Rashes, Low-grade fever, Weight loss, Dizziness, Dry eyes, Brittle hair, Stiffness, General unwell feeling
- Target variable: Diagnosis — 118 unique autoimmune disease labels

Two Classification Tasks:

- Task 1 — Binary: SLE vs Non-SLE (5,363 positive vs 39,842 negative)
- Task 2 — Binary: Endometriosis vs Non-Endometriosis (5,000 positive vs 40,205 negative)

4. Three/Four references to previous work (0.5 points)

4- References to Previous Work :

[1] A. Jorge *et al.*, "A fair machine learning model to predict flares of systemic lupus erythematosus," *JAMIA Open*, vol. 8, no. 4, p. ooaf072, 2025. [Online]. Available: https://doi.org/10.1093/jamiaopen/ooaf072

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[11] O. B. Fatade *et al.*, "Explainable AI for endometriosis diagnosis: A dense neural network approach with SHAP interpretation," *Int. J. Res. Sci. Innov.*, vol. 9, no. 4, pp. 123–130, 2025. [Online]. Available: https://doi.org/10.47772/IJRSI.2025.90412

[12] W. Wang *et al.*, "A stacked machine learning-based classification model for endometriosis and adenomyosis: A retrospective cohort study utilizing peripheral blood and coagulation markers," *Front. Digit. Health*, vol. 6, p. 1463419, Sep. 2024. [Online]. Available: https://doi.org/10.3389/fdgth.2024.1463419